Goshtasb Shahriari Mehr

Curriculum Vitae

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Education

Jan 2022 – Present **Ph.D.**, *Design, Construction & Planning (Urban & Regional Planning)*, *University of Florida,* Florida, United States — **defending Fall 2026**
Ph.D. Dissertation
Title: A Generative Agent-Based Model of Urban Food Access: Comparing Food-Access Interventions in a Data-Scarce Urban Food Desert

Aug 2023 – 2025 **M.Sc.**, *Electrical and Computer Engineering (Artificial Intelligence)*, *University of Florida,* Florida, United States — **Completed**,
GPA: 3.8/4.0

Sep 2016 – Aug 2019 **M.Sc.**, *Geographic Information Science Engineering*, *University of Tehran,* Tehran, Iran,
GPA: 16.58/20 (3.65/4)
M.Sc. Thesis
Title: Development of a recommender system using pattern recognition among users' points of interest and users' internet behavior
Supervisors: Prof. Mahmoud Reza Delavar, Prof. Christophe Claramunt
Advisor: Prof. Babak Nadjar Araabi
Grade: Excellent

Sep 2012 – Aug 2016 **B.Sc.**, *Geomatics Engineering*, *K.N. Toosi University of Technology,* Tehran, Iran,
GPA: 16.07/20 (3.27/4)
B.Sc. Thesis
Title: Evaluation of the methods of 3D Cadastre
Supervisor: Dr. Mohammad Karimi
Grade: 20/20

Certification

Jan 2023 – Dec 2024 **Graduate Certificate in Machine Learning**, Department of Electrical and Computer Engineering, University of Florida

Jan 2020 – Apr 2020 **Machine Learning** on Coursera

Mar 2020 **Deep Learning Specialization** on Coursera

- Mar 2020 ◦ **Neural Networks and Deep Learning** on Coursera
- May 2020 ◦ **Structuring Machine Learning Projects** on Coursera

Dec 2016 – Jan 2017 **Introduction to Programming with .NET** by Tehran Institute of Technology, Tehran, Iran

Dec 2016 – Jan 2017 **.NET Framework Application Development Foundation** by Tehran Institute of Technology, Tehran, Iran

Oct 2017 Workshop "**Empowering your Business with Web GIS and Mobile GIS**", ISPRS International Joint Conference, Tehran, Iran

Oct 2018 Workshop "**Teaching Assistant Training**", University of Tehran, Tehran, Iran

Research Experience

Jan 2022 – Present Research Assistant, University of Florida, Florida Institute for Built Environment Resilience (FIBER)

Jun 2023 – Present Research Assistant, University of Florida, Disasters, Trust, and Social Change (DTSC) Lab

Research Assistant, University of Florida, iAdapt Lab — comparative scenario analysis and public-transportation modeling for FDOT District 5 (Orlando, FL)

Sep 2017 – Aug 2019 Research Assistant, University of Tehran, Location Based Services

Selected Technical Projects

2026 **GeoChatBot** — embeddable, browser-native LLM agent for plain-English geospatial analysis with zero backend: plan-then-execute agent over 28 typed tools with a human approval gate, in-browser hybrid RAG (MiniLM embeddings + BM25 with Reciprocal Rank Fusion), DuckDB-WASM spatial SQL over Apache Arrow, MapLibre GL/deck.gl mapping; 800+ tests and an LLM-evaluation harness benchmarking 5+ models. Sole developer. *(Personal; public repo)*

2025 – 2026 **GeoMesa Food Access** (Ph.D. dissertation, DTSC Lab) — spatially-explicit rule-based and *generative* agent-based model of household food access in Health Zone 1, Jacksonville, FL: household agents decide via an LLM with two-tier RAG memory (episodic + social word-of-mouth); calibrated to census/BLS targets (0.095 calibration error), validated with Sobol sensitivity analysis and 50-seed Monte-Carlo replication; ~36,000 lines of Python, ~280 tests, dashboards deployed on Hugging Face Spaces.

2026 **Proxima** — mobile-first construction workforce-management SaaS built around an AI-assisted crew-scheduling optimizer (Google OR-Tools CP-SAT, generalized RCPSP) validated against published benchmarks at 0% optimality gap; React Native/Expo + FastAPI + Supabase multi-tenant architecture. Co-founder; UF Big Idea Competition semi-finalist. *(Venture)*

2026 **FieldSurvey** (DTSC Lab) — multi-tenant geospatial survey-analytics SaaS (~47,000 lines, solo): Next.js 15/React 19 front end, Supabase/PostGIS backend with row-level security, and a FastAPI spatial-statistics microservice (Getis-Ord Gi*, LISA, Kulldorff spatial scan, spatial regression) with permutation inference. *(Public repo)*

2026 **KeyStone Heights Research Platform** (DTSC Lab) — GIS community-health research platform over 11,300 parcels for a UF study: statewide cadastral ETL, 4-tier geocoding and record-linkage pipeline, composite risk-score modeling, LLM map-driving agent; deployed and security-audited.

2026 **Local Glean** (DTSC Lab) — private AI email assistant: 9-node LangGraph drafting pipeline with fail-closed verification and a two-stage classifier (vector kNN cascading to an LLM) at 96.7% accuracy, learning from user corrections without retraining.

2026 **Cedar Key Bayesian Re-Analysis** (DTSC Lab) — replaced 195 univariate regressions with a seven-model Bayesian and causal portfolio (graded-response IRT, latent class analysis, SEM mediation, Bayesian Causal Forests, DAG discovery) in PyMC, documented with a pre-publication forensic audit.

2025 – 2026 **PetroConnect** — B2B industrial marketplace (∼125,000 lines; Next.js 15, PostgreSQL/Prisma, Docker) with a double-entry wallet ledger and idempotent payment processing. Co-founder & CTO. *(Venture)*

Selected Course Projects

Fall 2024 **Optimizing Built Environment Feature Detection Using Transfer Learning on CIFAR-100**, Pattern Recognition

Spring 2024 **Flower Species Classification and Lung Segmentation from X-ray Images**, Applied Machine Learning

Fall 2023 **Implementing CNN and MLP on Kuzushiji-MNIST**, Neural Networks & Deep Learning

Fall 2022 **Handwritten Mathematical Symbols Classification**, Fundamentals of Machine Learning

Spring 2017 **Landfill Site Selection Using Genetic Algorithms**, Intelligent Computational Methods

Fall 2014 **Urban Hospital Site Selection Combining MCDM and GIS Spatial Analysis**, Fundamentals of Urban Management & Planning

Honors And Awards

2026 Semi-finalist, UF Big Idea Competition, University of Florida (Proxima)

2021 Graduate Support Funding award from the Provost office at the University of Florida

2016 Ranked 10th among 2,000 participants of the national graduate entrance exam in Iran (ranked 1st among all K.N. Toosi University participants)

2012 Ranked in the top 1% of the approximately 300,000 participants of the national university entrance exam in Iran

Research Interests

- Urban Resilience & Food Access
- Geospatial AI & Spatial Data Science
- Agent-Based & Generative-Agent Modeling
- LLM Agents & Retrieval-Augmented Generation
- Location-Based Services
- Spatio-temporal Analysis
- Urban Analytics
- Mobility Analysis

Academic Experience

Fall 2023 Teaching Assistant, University of Florida, "Introduction to Urban Analytics"

Fall 2023 Teaching Assistant, University of Florida, "Introduction to Planning Information Systems"

2017 – 2019 Teaching Assistant, University of Tehran, "Geographic Information Systems I" (GIS/ArcGIS, 3 semesters)

Technical Skills

Programming	Python, TypeScript, JavaScript, SQL, Swift/SwiftUI, Bash, C#, MATLAB, LaTeX
AI / ML	LLM agents & RAG, LangGraph, prompt engineering, embeddings & vector search (pgvector, BM25), PyTorch, scikit-learn, deep learning (CNN, transfer learning), LLM evaluation & safety
Statistics & Simulation	Bayesian inference (PyMC/MCMC), causal inference, agent-based modeling (Mesa/Mesa-Geo), discrete-choice models, Sobol sensitivity analysis, Monte-Carlo, optimization (OR-Tools CP-SAT)
Geospatial	GeoPandas, Shapely, PostGIS, ArcGIS, spatial statistics (Moran's I, Getis-Ord, LISA), geocoding & record linkage, MapLibre GL, deck.gl, Leaflet, DuckDB spatial, ENVI
Web & Cloud	Next.js/React, React Native/Expo, FastAPI, Django, Docker, Google Cloud Run, Vercel, Supabase, Hugging Face Spaces, GitHub Actions, pytest/Vitest/Playwright
Databases	PostgreSQL/PostGIS, pgvector, DuckDB, SQLite, MySQL, Oracle

Publications

2024 N. Soltani Nejad, R. Rastegar, G. Shahriari Mehr, F. Taheri Azad, "**Conceptualizing Tourist Journey: Qualitative Analysis of Tourist Experiences on TripAdvisor**," *Journal of Quality Assurance in Hospitality and Tourism*

2021 G. Shahriari Mehr, M. Delavar, C. Claramunt, B. N. Araabi, M. R. A. Dehaqani, "**A Store Location-based Recommender System Based On User's Position and Web Searches**," *Journal of Location Based Services*. doi.org/10.1080/17489725.2021.1880029

2019 G. Shahriari Mehr, M. Delavar, C. Claramunt, B. N. Araabi, M. R. A. Dehaqani, "**Discover Points of Interest based on Users' Internet Searches Through An Online Shopping Website**," presented at the ISPRS International Joint Conference 2019

Manuscripts in Preparation

G. Shahriari Mehr et al., "A literature-calibrated spatial agent-based model for comparing food-access interventions in a data-scarce urban food desert: a case study of Health Zone 1, Jacksonville, FL"

G. Shahriari Mehr et al., generative agent-based modeling of urban food access (dissertation core)